

YSTECH HIGH FREQUENCY SWITCH IBDC10050 & IBDC30060 DCDC MODULE CAN bus communication protocol

User 's manual



Contact us.

WeChat: +86 13559210897

WhatsApp: +86 13559210897

Email: yewsource@hotmail.com



WeChat



WhatsApp

YSTECH

[illegible]

1 Normative description

This protocol specifies the CAN communication protocol between bidirectional isolated power modules (IBG series and IBDC series) and STS modules (GSTS series) and the EMS (Energy Management System).

The CAN identifier used in this protocol is 29-bit extended frame, and the communication baud rate can be set to 1000k, 500k, 250k, 125kpbs, and the default is 125kpbs. The data transmission of this protocol adopts the big-end mode, i.e., send the high bit first, and then send the low bit (e.g., 0x1234, send 0x12 first, and then send 0x34). (e.g. 0x1234, send 0x12 first, then send 0x34).

Product models supported by this protocol:

serial number	Product Model	Product Description
1	IBDC10050	DCDC Module
2	IBDC30060	DCDC Module

2 Network Topology

The CAN bus interconnection structure is shown in Fig. 1. CAN BUS is a bus for data communication between several network nodes. There are several module nodes and an EMS on a CAN BUS, and the EMS obtains the data and status of each power module through CAN communication, and at the same time, it can control the working mode and operation status of the power module.

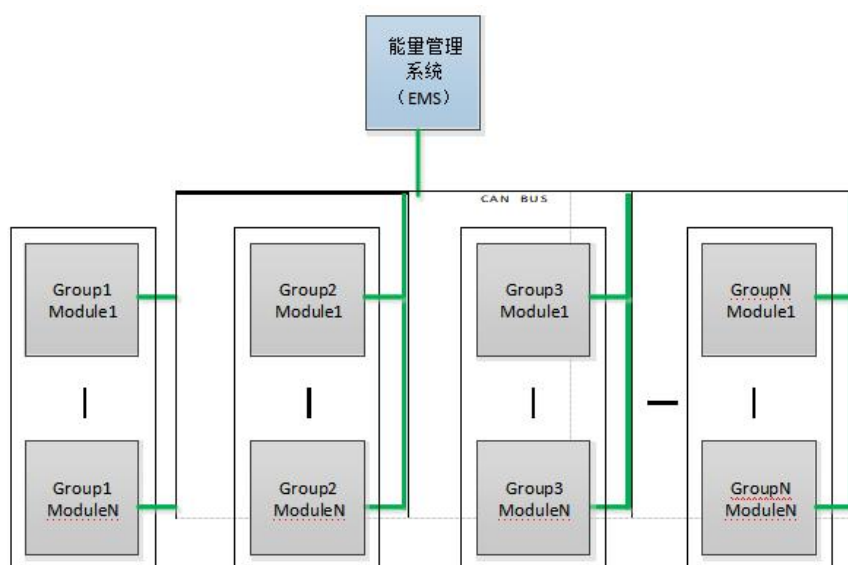


Figure 1 CAN-BUS network topology

3 CAN Data Frame Format

The CAN data frame consists of a 29-bit identifier field and a 64-bit data field.

Table 1 CAN data frame format

identifiers	data domain			
Bit0 - Bit28	Byte 1	Byte 2	...	Byte 8
CAN ID	Data (Byte1 - Byte8)			

3.1 CAN Identifiers

The actual definition of the 29-bit identifier using the CAN extended frame is shown in Table 2.

Table 2 CAN Extended Frame Identifier Definitions

28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRIO			R	DP	PF								PS								SA							
Reserve (5bit) reservations				FTP (8bit) Frame Type								MID (8bit) module address								GID(5bit) Module Group Address				MNT(3bit) Monitor Address				
particular																												
PRIO				Frame Priority, supports 8 priorities, default is 0.																								
R				Reserved bits, fixed to 0																								
DP				Reserved bits, fixed to 0																								
PF				FTP (Frame Type, bit16 - bit23) -----Frame Type: 0x74: Query frame 0x73: Setting the frame 0x72/0xF2: Query or set response frame																								
PS				MID (Module ID, bit8 - bit15) --- module address The module address consists of the hardware address HW_Add ^[1] and the offset address Offset_Add ^[2] : PS = MID ^[3] = HW Add + Offset Add																								
SA				GID (Group ID, bit3 - bit7) --- group address After the modules have been grouped, the address set for each group is in the range 1-31, default is 1, where 31 is the global broadcast ID ^[4] (all groups can receive it, only used for setting frames). MNT(Monitor, bit0 - bit3) - Monitor address (controller, EMS, etc.) The range is 0-7, and the default is 7. SA ^[5] = GID *8 + MNT																								

The [1] module has a 5-bit dial code that defines 32 hardware addresses ($2^5 = 32$) in the range 0x00 - 0x1F, where 0x1F is the in-group broadcast address.

[2] Offset address is used to distinguish different models of modules to avoid duplication of address when the modules are connected to the same bus, the definition of offset address is shown in Table 3:

Table 3 Module Offset Address Definitions

serial number	Equipment Model	offset address
1	IBDC10050	0x60
2	IBDC30060	0x80

[3] The list of module address MIDs is shown in Table 4:

Table 4 Module Address List

module (in software)	PS		
Module Model	MID	Offset_Add + HW_Add	Remarks
IBDC10050	module address	0x60 + hardware address	Module address value [0x60,0x7E], default is 0x60
	in-group broadcast address	0x60 + 0x1F	Fixed to 0x7F
IBDC30060	module address	0x80 + hardware address	Module address value [0x80,0x9E], default 0x80
	in-group broadcast address	0x80 + 0x1F	Fixed to 0x9F

[4] Broadcasts are categorized as intra-group and global broadcasts.

The effective range of intra-group broadcast is all the modules in the same group, and the module receives the intra-group broadcast and responds to it;

The effective range of the global broadcast is all modules within all groups, and modules do not answer after receiving the global broadcast;

Global broadcasts have a higher priority than intra-group broadcasts. For the same frame, the module parses global broadcasts first, then intra-group broadcasts;

Broadcast is only used for single frame setup frames.

[5] The module shall answer when it receives a query frame or a setup frame (including intra-group broadcasts) from the monitor. The PS and SA of the received frame shall be exchanged and the PF shall be changed to a response frame.

3.2 Data fields

The length of the data field is 64 bits (8 bytes). When the length of a message is greater than 8 bytes, the sender needs to split the message into several CAN data frames. If the data field is less than 8 bytes, use 0x00 to make up the 8 bytes. The protocol defines the 64-bit data field as four 16-bit words: WORD0, WORD1, WORD2, WORD3, as shown in the table below:

64bit data field							
Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8
Word0		Word1		Word2		Word3	

3.3 CAN Data Frames

A complete data frame defined by this protocol is shown below:

Request frame format:

CANID			WORD0	WORD1	WORD2	WORD3
0x73/0x74	PS	SA	Order	Data0	Data1	Data2

Response frame format:

CANID			WORD0	WORD1	WORD2	WORD3
0x72/0x22/0xF2	SA	PS	Order	Data0	Data1	Data2

Note: Order indicates the frame number of each frame of data defined in the module data table.

3.3.1 Query Frames

CANID			WORD0	WORD1	WORD2	WORD3
0x74	PS	SA	Order	Start Order	Len	0

When Order=0, it means to query multi-frame data, in this case, Start Order means the starting frame number and Len means the number of query. When Order>0, it means querying single frame data, at this time, Order indicates the frame number to be queried, and Start Order and Len are both 0.

Note: A query frame can only read the contents of the frame data bit readable (R or RW).

3.3.2 Query Response Frames

Single-frame query response

CANID			WORD0	WORD1	WORD2	WORD3
0xF2	SA	PS	Order	Data0	Data1	Data2

Order indicates the serial number of the queried data frame, the data are Data0, Data1, Data2.

Example:

Monitor → Module: 0x0074600F 00 03 00 00 00 00 00 00 00

SA	0x0F	GID = 1, MNT = 7
PS	0x60	Offset_Add = 0x60, HW_Add = 0
PF	0x74	query frame
Order	0x0003	>0 for query single frame
Start Order	0x0000	Fixed to 0
Len	0x0000	Fixed to 0
analyze	Monitor query for frame 3 data for module group 1, module type 0, IBDC10050	

Module → Monitor: 0x00F20F60 00 03 00 03 01 23 04 56

SA	0x60	Offset_Add = 0x60, HW_Add = 0
PS	0x0F	GID = 1, MNT = 7
PF	0xF2	Query Response Frame
Order	0x0003	The frame number is 3
Data0	0x0003	"Battery state of charge" is "3 - constant voltage"
Data1	0x0123	"Charging power" is 291W.
Data2	0x0456	"Discharge power" is 1110W.
analyze	Frame 3 data from Module Response for Module Group 1, Module Type 0, IBDC10050	

multiframe query response

Frame 1

CANID			WORD0	WORD1	WORD2	WORD3
0xF2	SA	PS	1st Order	Data0	Data1	Data2

Frame 2

CANID			WORD0	WORD1	WORD2	WORD3
0xF2	SA	PS	2nd Order	Data0	Data1	Data2

...

Frame Len

CANID			WORD0	WORD1	WORD2	WORD3
0xF2	SA	PS	Last Order	Data0	Data1	Data2

Order indicates the frame number to be queried, the data are Data0, Data1, Data2.

Example:

Monitor → Module: 0x0074610F 00 00 00 03 00 06 00 00

SA	0x0F	GID = 1, MNT = 7
PS	0x61	Offset_Add = 0x60 , HW_Add = 1
PF	0x74	query frame
Order	0x0000	= 0 for multi-frame lookup
Start Order	0x0003	Starting frame number 3
Len	0x0006	Query 6 consecutive frames of data
analyze	Monitor query frame 3 - frame 8 data for module group 1, module type IBDC10050	

Module → 0x00F20F61 00 03 00 03 01 23 04 56

Monitor: 0x00F20F61 00 04 01 E5 04 1A 00 00

0x00F20F61 00 05 00 00 00 00 00 00

0x00F20F61 00 06 00 00 00 00 00 00

0x00F20F61 00 07 00 00 00 00 00 00

0x00F20F61 00 08 00 00 01 31 00 00

First Order:

SA	0x61	Offset_Add = 0x60 , HW_Add = 1
PS	0x0F	GID = 1, MNT = 7
PF	0xF2	Query Response Frame
Order	0x0003	The frame number is 3
Data0	0x0003	"Battery state of charge" is "constant voltage".
Data1	0x0123	"Charging power" is 291W.
Data2	0x0456	"Discharge power" is 1110W.
analyze	Frame 3 data from the module response of Module Group 1, Module Type IBDC10050	

Second Order.

SA	0x61	Offset_Add = 0x60 , HW_Add = 1
PS	0x0F	GID = 1, MNT = 7
PF	0xF2	Query Response Frame
Order	0x0004	The frame number is 4
Data0	0x01E5	" Battery voltage " is 48.5V
Data1	0x041A	" Battery current " is 105A

Data2	0x0000	"RSVD" is 0
analyze	Frame 4 data of the module response for Group 1, Module #1, Module Type IBDC10050	

...

Last Order.

SA	0x61	Offset_Add = 0x60 , HW_Add = 1
PS	0x0F	GID = 1, MNT = 7
PF	0xF2	Query Response Frame
Order	0x0008	The frame serial number is 8
Data0	0x0000	"RSVD" is 0.
Data1	0x0131	"Ambient Temperature" is 30.5° C.
Data2	0x0000	"RSVD" is 0.
analyze	Frame 8 data of the module response for Group 1, Module #1, Module Type IBDC10050	

3.3.3 Setting/control frames

Request Frame:

CANID			WORD0	WORD1	WORD2	WORD3
0x73	PS	SA	Order	Set Value	0	0

Order indicates the frame number to be set/controlled, Set Value indicates the amount to be set/controlled, and Data1 and Data2 are 0.

Response Frame:

CANID			WORD0	WORD1	WORD2	WORD3
0x22/0x72	SA	PS	Order	Set Value	0	0

Note: Setup/Control frames only support writing 1 parameter.

Example:

1, point to point setting

Monitor → Module: 0x0073600F 00 36 02 58 00 00 00 00 00

SA	0x0F	GID = 1, MNT = 7
PS	0x60	Offset_Add = 0x60, HW_Add = 0
PF	0x73	Setting the frame
Order	0x0036	Setting of frame 54 parameter "Battery charging voltage".
Set Value	0x0258	Set value: 60V (0x0258 = 600) 600*0.1 = 60V

analyze	The monitor sets the frame 54 parameter value of module 0 in group 1 to 0x0258 and the module type to IBDC10050.
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Module → Monitor: 0x00220F60 00 36 02 58 00 00 00 00 00

SA	0x60	Offset_Add = 0x60, HW_Add = 0
PS	0x0F	GID = 1, MNT = 7
PF	0x22	Setting the response frame
Order	0x0036	Frame 54 Parameter "Battery equalization voltage"
Set Value	0x0258	Current Value: 60V (0x0258 = 600) 600*0.1 = 60V
analyze	Module Group 1, Module 0, Frame 54 parameterization was successful. Module type is IBDC10050	

2. Group broadcast settings

Monitor → Module: 0x00737F0F 00 37 01 90 00 00 00 00 00

SA	0x0F	GID = 1, MNT = 7
PS	0x7F	Offset_Add = 0x60, HW_Add = 1F (in-group broadcast)

PF	0x73	Setting the frame
Order	0x0037	Setting of frame 55 parameter "Charging current".
Set Value	0x0190	Set value: 40A (0x0190 = 400) 400*0.1 =40A
analyze	Monitor sets the Frame 55 parameter value to 0x190 for all modules in Group 1 and the module type to IBDC10050.	

Module →

Monitor:

0x00220F60 00 37 01 90 00 00 00 00

0x00220F61 00 37 01 90 00 00 00 00

0x00220F62 00 37 01 90 00 00 00 00

0x00220F63 00 37 01 90 00 00 00 00

[Assuming the first group has 4 modules (module hardware addresses 0-3), all modules respond]

Module 0:

SA	0x60	Offset_Add = 0x60 , HW_Add = 0
PS	0x0F	GID = 1 , MNT = 7
PF	0x22	Setting the response frame
Order	0x0037	Frame 55 Parameter "Charge current"
Set Value	0x0190	Current value: 40A (0x0190 = 400) $400 \times 0.1 = 40A$
analyze	Module Group 1, Module 0, Frame 55 parameterization was successful. Module type is IBDC10050	

Module 1

SA	0x61	Offset_Add = 0x61 , HW_Add = 0
PS	0x0F	GID = 1 , MNT = 7
PF	0x22	Setting the response frame
Order	0x0037	Frame 55 Parameter "Charge current"
Set Value	0x0190	Current value: 40A (0x0190 = 400) $400 \times 0.1 = 40A$
analyze	The parameters of frame 55 of module group 1, module No. 1, have been set successfully. Module type is IBDC10050	

Module 2:

SA	0x62	Offset_Add = 0x61 , HW_Add = 0
PS	0x0F	GID = 1 , MNT = 7
PF	0x22	Setting the response frame
Order	0x0037	Frame 55 Parameter "Charge current"
Set Value	0x0190	Current value: 40A (0x0190 = 400) $400 \times 0.1 = 40A$
analyze	Module Group 1, Module 2, Frame 55 parameterization was successful. Module type is IBDC10050	

Module 3:

SA	0x63	Offset_Add = 0x61 , HW_Add = 0
PS	0x0F	GID = 1 , MNT = 7
PF	0x22	Setting the response frame
Order	0x0037	Frame 55 Parameter "Charge current"

Set Value	0x0190	Current value: 40A (0x0190 = 400) 400*0.1=40A
analyze	Module Group 1, Module 3, Frame 55 parameterization was successful. Module type is IBDC10050	

3. Global broadcast settings

Monitor → Module: 0x00737FFF 00 20 00 AA 00 00 00 00 00 00

SA	0xFF	GID = 31 (global broadcast) , MNT = 7
PS	0x7F	Offset_Add = 0x60 , HW_Add = 1F (in-group broadcast)
PF	0x73	Setting the frame
Order	0x0020	Setting of frame 32 parameter "System On/Off"
Set Value	0x00AA	Set value: 170 (off)
analyze	Monitor sets the Frame 32 parameter value to 170 for all groups of modules with module type IBDC10050.	

Attention:

- 1. Global broadcast setting, the module only receives the execution, and does not respond;
- 2. When both global broadcast request and intra-group broadcast request exist in the same setup frame, all modules prioritize the global broadcast request.

4 Data sheets

4.1 Communication Interface Definitions

- Baud rate: 125kbps (default), optional 125k, 250k, 500k, 1000k;
- The CAN interface design varies from module to module, so please refer to the actual situation.

4.2 Data sheets

4.2.1 Data sheet description:

- Order must be unique and non-zero in the same data table;
- The three data coefficients and data types in the same Order must be consistent;
- Actual = Data * Coefficient;

RSVD Indicates unused data, which is temporarily retained, and the read response is 0, so writing is prohibited;

:: Data attributes are categorized as read-only (R), write-only (W), and read/write (R/W)

- The data type USHORT indicates a 16-bit unsigned number;
- The data type SHORT represents a 16-bit signed number.
- Unless otherwise specified, the units of common data are defined as follows:

Voltage values are in V (volts);

Current values are in A (amps);

Temperature values are in ° C (degrees Celsius);

Frequency values are in Hz (hertz);

Active power values are in W (watts);

Reactive power values in VAR (spent)

Apparent power values in VA (volt-amperes)

- For details of the fault alarm, refer to the attached table.
- This protocol contains both DCDC and DCAC data, so please be careful to distinguish them.

4.2.2 Data sheets

Refer to the XXXXXX Power Module CAN Data Sheet (User Edition)

5 Examples of common commands and operations

This section lists the commonly used request/response frames and procedures for reference, using the Group 1, No. 0 IBDC10050 module as an example.

[GID = 1, Offset_Add = 0x60, HW_Add = 0]

I. Setting the basic parameters of the module

1. Set the working mode:

	00 34 00 00 00 00 00 00
(1) Set	00 34 00 00 00 00 00 00
charge/discharge	
command - standby	
Request 0x0073600F	
Response	00 34 00 01 00 00 00 00
0x00220F60	00 34 00 01 00 00 00 00

(2) Set	
charge/discharge	00 34 00 02 00 00 00 00
command - charge	00 34 00 02 00 00 00 00
Request 0x0073600F	
Response	
0x00220F60	

(3) Setting	
charge/discharge	
instructions -	
Discharge Request	
0x0073600F	
Response	
0x00220F60	

(4) Setting charging and discharging commands - BAT automatic switching	
request 0x0073600F	00 34 00 03 00 00 00 00
ing 0x00220F60	00 34 00 03 00 00 00 00
respons	
ive	

(5) Setting charging and discharging instructions - BUS automatic switching	
request 0x0073600F	00 34 00 04 00 00 00 00
ing 0x00220F60	00 34 00 04 00 00 00 00
respons	
ive	

2. Set the battery parameters:

(1) Setting the battery type - lead-acid battery

Request 0x0073600F 00 30 00 00 00 00 00 00 00 00

Response 0x00220F60 00 30 00 00 00 00 00 00 00 00

(2) Setting the

battery type- 00 30 00 01 00 00 00 00

Lithium Battery 00 30 00 01 00 00 00 00

Request 0x0073600F

Response

0x00220F60

(3) Set equalizing voltage - 60V ($0x0258 = 600$) $600 \times 0.1 = 60V$

Request `0x0073600F 00 36 02 58 00 00 00 00 00`

Response `0x00220F60 00 36 02 58 00 00 00 00 00`

(4) Set float voltage - 60V ($0x0258 = 600$) $600 \times 0.1 = 60V$

Request `0x0073600F 00 38 02 58 00 00 00 00 00`

Response `0x00220F60 00 38 02 58 00 00 00 00 00`

(5) Set charge current - 50A ($0x01F4 = 500$) $500 \times 0.1 = 50A$

Request `0x0073600F 00 37 01 F4 00 00 00 00 00`

Response `0x00220F60 00 37 01 F4 00 00 00 00 00`

(6) Set discharge voltage - 40V ($0x0190 = 400$) $400 \times 0.1 = 40V$

Request `0x0073600F 00 39 01 90 00 00 00 00 00`

Response `0x00220F60 00 39 01 90 00 00 00 00 00`

(7) Set discharge current - 50A ($0x01F4 = 500$) $500 \times 0.1 = 50A$

Request `0x0073600F 00 3A 01 F4 00 00 00 00 00`

Response `0x00220F60 00 3A 01 F4 00 00 00 00 00`

(8) Setting charging and discharging power limit percentage - 100.0% ($0x0064 = 100$)

Request `0x0073600F 00 35 03 E8 00 00 00 00 00`

Response `0x00220F60 00 35 03 E8 00 00 00 00 00`

3. Set the battery protection parameters:

(1) Set battery overvoltage protection point - 110V ($0x044C = 1100$) $1100 \times 0.1 = 110V$

Request `0x0073600F 00 3B 04 4C 00 00 00 00 00`

Response `0x00220F60 00 3B 04 4C 00 00 00 00 00`

(2) Set battery low voltage alarm point - 38V ($0x017C = 380$) $380 \times 0.1 = 38V$

Request `0x0073600F 00 3C 01 7C 00 00 00 00 00`

Response `0x00220F60 00 3C 01 7C 00 00 00 00 00`

(3) Set battery low voltage protection point - 35V ($0x015E = 350$) $350 \times 0.1 = 35V$

Request `0x0073600F 00 3D 01 5E 00 00 00 00 00`

Response `0x00220F60 00 3D 01 5E 00 00 00 00 00`

4. Module switching operation.

- | | | | |
|--------|---------|-----------------------------|--|
| (1) | -up | responsive | |
| Module | request | | |
| enable | stop | (2) Module shutdown request | |
| power | | responsive | |

0x0073600F	00 20 00 55
0x00220F60	00 00 00 00
	00
	00 20 00 55
	00 00 00 00
	00
0x0073600F	
0x00220F60	
	00 20 00 AA
	00 00 00 00
	00 20 00 AA
	00 00 00 00

II. Basic information on the query module

1. Query DCDC data frames 2 - 14:

```

reque  0x0074600F  00 00 00 02 00 0D 00 00 (order=0 startorder = 2 orderlen = 13)
sting
respon  0x00F20F60  00 02 00 00 00 00 00 00
sive
respon  0x00F20F60  00 03 00 00 00 00 00 00
sive
respon  0x00F20F60  00 04 03 d4 00 00 00 00
sive
respon  0x00F20F60  00 00 00 00 00 00 00 00
sive
respon  0x00F20F60  00 06 00 00 00 00 00 00
sive
respon  0x00F20F60  00 07 00 00 00 00 00 00
sive
respon  0x00F20F60  00 08 00 00 00 FA 00 00
sive
respon  0x00F20F60  00 09 00 00 00 00 00 00
sive
respon  0x00F20F60  00 0A 00 00 00 00 00 00
sive
respon  0x00F20F60  00 0B 00 00 00 00 00 00
sive
respon  0x00F20F60  00 0C 00 00 00 00 00 00
sive
respon  0x00F20F60  00 0D 00 00 00 00 00 00
sive
respon  0x00F20F60  00 0E 00 00 00 00 00 00
sive

```